RNA-seq, WSI, and WSI–RNA Fusion for PAM50 Subtype Classification on TCGA-BRCA

1 Problem Definition

We evaluate WSI, RNA-seq, and WSI–RNA fusion branches for four-class PAM50 molecular subtype classification on the TCGA-BRCA cohort. The task is slide-level: each H&E whole-slide image is classified into Luminal A, Luminal B, Basal-like, or HER2-enriched. The RNA branch acts as a molecular reference and as the second modality in fusion. Normal-like slides and slides without matched RNA-seq are excluded so that every reported model is trained and evaluated on the same cohort.

Table 1: PAM50 classes used in this report.

Class index	Label	Description
0	Luminal A	Hormone receptor-positive, low proliferation
1	Luminal B	Hormone receptor-positive, higher proliferation
2	Basal-like	Triple-negative, aggressive
3	HER2-enriched	HER2 overexpression

All experiments in this report are run on the matched cohort of **986** slides with both UNI2-h WSI features and RNA-seq features. The class distribution is given in Table 2.

Table 2: Matched WSI–RNA cohort used for every reported experiment.

Class index	PAM50 subtype	Slides
0	Luminal A	508
1	Luminal B	222
2	Basal-like	173
3	HER2-enriched	83
	Total	986

2 Pipeline

The pipeline trains an RNA branch and a WSI branch on the matched cohort, then combines them with several fusion strategies that share the same patient-level splits.

2.1 Stage 1: RNA-seq Preprocessing

RNA-seq counts were filtered to the four PAM50 classes, joined with the WSI cohort by TCGA case identifier, and converted into CLAM-compatible sample tables. Within each training fold, the 10,000 highest-variance genes are selected and standardised using training-fold mean and standard deviation. The selected transform is then applied to validation and test samples. Gene selection and scaling are refit per fold so that test-fold RNA distributions cannot leak into training.

2.2 Stage 2: RNA Branch

The RNA branch is a multilayer perceptron that maps the standardised gene vector to four logits. The configuration was chosen by a Bayesian sweep on fold-0 validation AUC and is summarised in Table 3.

Table 3: RNA branch configuration selected by the fold-0 validation sweep.

Parameter	Value
Architecture	MLP
Hidden dimensions	[1024, 512]
Input genes	10,000 top-variance
Dropout	0.4
Optimizer	AdamW
Learning rate	1.97e–4
Weight decay	7.01e–4
Label smoothing	0.05
Weighted loss / sampling	Enabled
Batch size	64
Max epochs	80
Early stopping patience	16

2.3 Stage 3: WSI Branch

The WSI branch is CLAM-MB on UNI2-h slide features. Each slide is encoded as a bag of patch embeddings extracted with the UNI2-h foundation model and aggregated by CLAM attention. The model predicts PAM50 subtype from H&E morphology alone.

2.4 Stage 4: WSI–RNA Fusion

Each fusion model pairs the CLAM-MB WSI branch with an RNA MLP branch. Fold splits and the per-fold RNA transform are reused across all fusion variants:

- **Concat fusion**: concatenate the CLAM slide embedding and the RNA embedding, then classify through a fusion head with hidden dimension 32.

- **Gated fusion**: project WSI and RNA to a shared 32-dimensional space, learn a sigmoid gate from the joint embedding, and classify the gated mixture.
- **Late probability fusion**: linearly combine RNA and WSI class probabilities with a per-fold convex weight selected on the validation split.
- **Residual fusion**: keep the RNA prediction as the base and train a small WSI-conditioned head that produces a conservative residual correction.
- **Selective ensemble**: temperature-calibrate RNA, WSI, concat, gated, and residual probabilities on the validation split, then combine them with fold-specific convex weights selected by validation balanced accuracy.

3 Experimental Setup

3.1 Hyperparameter Optimisation

The RNA configuration in Table 3 was selected by a Weights & Biases Bayesian sweep that maximised fold-0 validation AUC. The WSI branch reuses the existing CLAM-MB configuration. Concat, gated, and residual fusion heads were chosen by validation-only sweeps on fold 0; the residual sweep also fixes a residual scale of 0.1, fusion hidden dim 16, dropout 0.25, learning rate 1.65e–4, and weight decay 6.91e–6.

3.2 Evaluation Protocol

Every model uses the same 10-fold patient-level split. Each patient appears in exactly one held-out test fold, so concatenating the ten test sets gives one held-out prediction for every slide in the 986-slide cohort. Hyperparameters are selected on validation data only, and test metrics are reported after the validation choices are fixed. The primary metrics are macro AUC, accuracy, balanced accuracy, macro F1, and weighted F1. Macro AUC is one-vs-rest and is averaged across classes; balanced accuracy and macro F1 are reported because the class distribution is imbalanced.

4 Results

4.1 Aggregate Held-out Test Performance

Table 4 reports metrics after concatenating held-out test predictions across all ten folds. Best values per column are bolded.

Table 4: Aggregate held-out test results on the matched 986-slide cohort.

Method	n	Macro AUC ↑	Acc. ↑	Bal. Acc. ↑	Macro F1 ↑	Weighted F1 ↑	ECE ↓
WSI only	986	0.8879	0.7181	–	0.65	0.7179	–
Matched RNA only	986	0.9785	0.8783	0.8809	0.8644	0.8799	–
WSI + RNA concat	986	0.9765	0.8884	0.8745	0.8653	0.8896	–
WSI + RNA gated	986	0.9744	0.8874	0.8724	0.8667	0.8885	–
Late probability fusion	986	0.9789	0.8803	0.8756	0.8657	0.8812	–
WSI + RNA residual	986	0.9787	0.8803	0.8535	0.8575	0.8797	–
Selective ensemble	986	**0.9796**	**0.8996**	**0.8856**	**0.8804**	**0.8999**	0.0256

The matched RNA-only branch is the strongest single-modality model, which matches the fact that PAM50 is defined from gene-expression patterns. WSI alone trails RNA by roughly 16 accuracy points and 22 macro F1 points. Direct feature-level fusion (concat, gated) reaches the RNA-only accuracy and slightly improves it, but does not improve macro AUC over RNA. Late probability fusion gives the best aggregate macro AUC among non-ensemble methods (0.9789), and residual fusion stays close to the RNA baseline. The **selective ensemble is the strongest model on every reported metric**: 0.9796 macro AUC, 0.8996 accuracy, 0.8856 balanced accuracy, 0.8804 macro F1, 0.8999 weighted F1, and 0.0256 expected calibration error.

4.2 Fold-level Stability

Table 5 reports mean and standard deviation across the ten held-out test folds. Aggregate metrics in Table 4 differ slightly from fold means because they are computed after concatenating all held-out predictions.

Table 5: Fold-level held-out test performance on the matched 986-slide cohort.

Method	Test macro AUC ↑	Test accuracy ↑
Matched RNA only	0.9806 ± 0.0075	0.8787 ± 0.0285
WSI + RNA concat	0.9783 ± 0.0062	0.889 ± 0.0276
WSI + RNA gated	0.9751 ± 0.0065	0.8879 ± 0.0282
Late probability fusion	0.9806 ± 0.0067	0.8806 ± 0.0268
WSI + RNA residual	0.9799 ± 0.009	0.8804 ± 0.0286
Selective ensemble	$\mathbf{0.9829 \pm 0.0053}$	$\mathbf{0.9001 \pm 0.0248}$

The selective ensemble has the highest fold-level mean and the lowest standard deviation on both AUC and accuracy. Per-fold AUC for the matched RNA branch and the selective ensemble is shown in Figure 1.

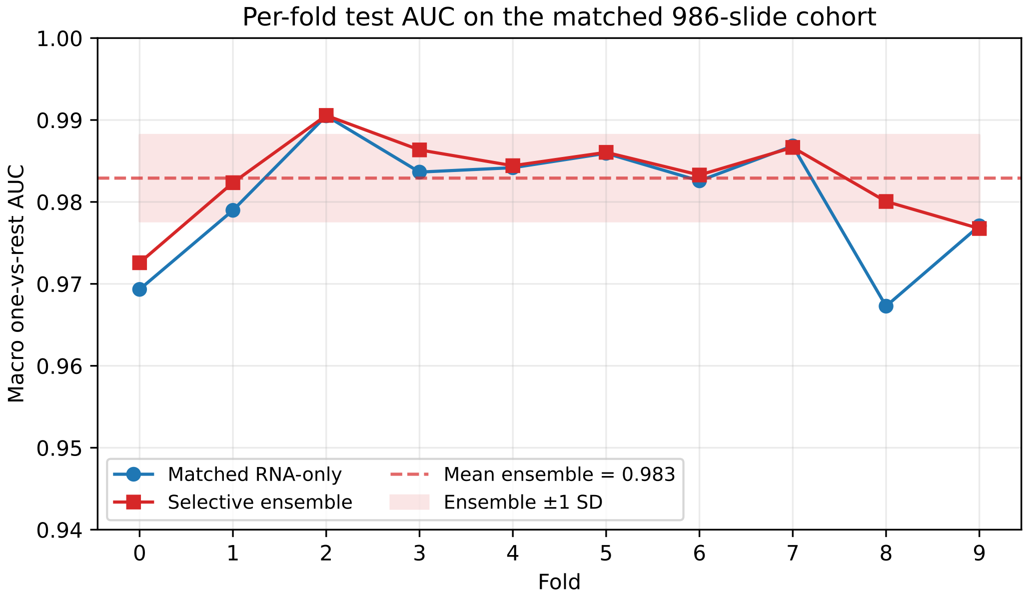

Figure 1: Per-fold held-out test AUC for the matched RNA branch and the selective ensemble on the 986-slide cohort. The dashed line and shaded band show the ensemble mean and one standard deviation.

4.3 Selective Ensemble Per-class Behaviour

Per-class metrics for the best aggregate model are in Table 6.

Table 6: Selective ensemble per-class held-out test metrics on the 986-slide cohort.

Class	AUC ↑	Precision ↑	Recall ↑	F1 ↑	Support
Luminal A	0.9765	0.9306	0.9232	0.9269	508
Luminal B	0.9604	0.8145	0.8108	0.8126	222
Basal-like	**0.9986**	**0.9826**	0.9769	**0.9797**	173
HER2-enriched	0.9829	0.7753	**0.8313**	0.8023	83

Basal-like is essentially separable from the other three classes; the remaining errors are concentrated in the Luminal A / Luminal B border and in the smaller HER2-enriched class. The confusion matrix in Figure 2 shows the same pattern: 53 Luminal A slides are misclassified as Luminal B and 32 in the opposite direction, while Basal-like reaches 0.98 row recall.

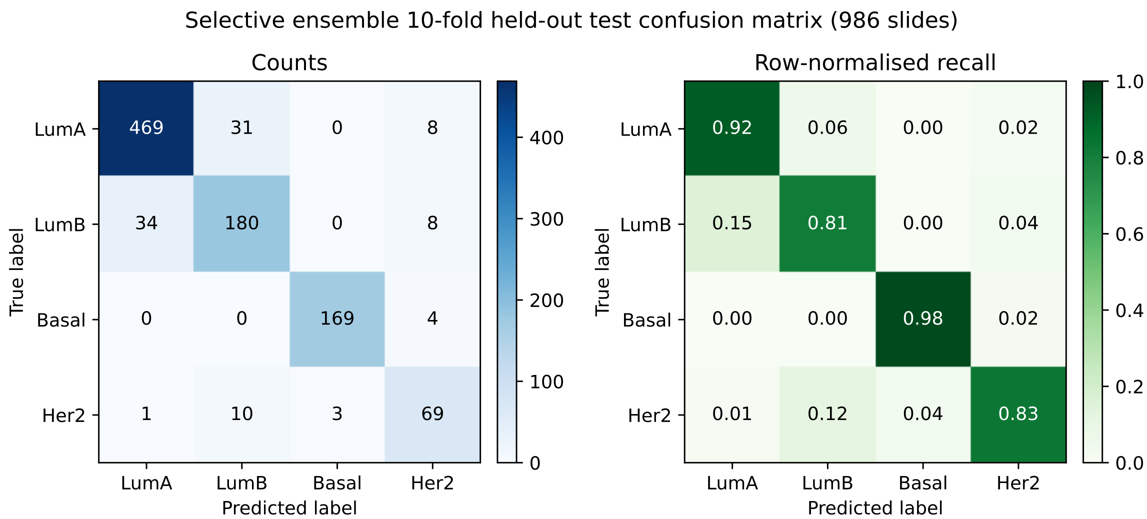

Figure 2: Selective ensemble 10-fold held-out test confusion matrix on the 986-slide cohort. Left: raw counts. Right: row-normalised recall.

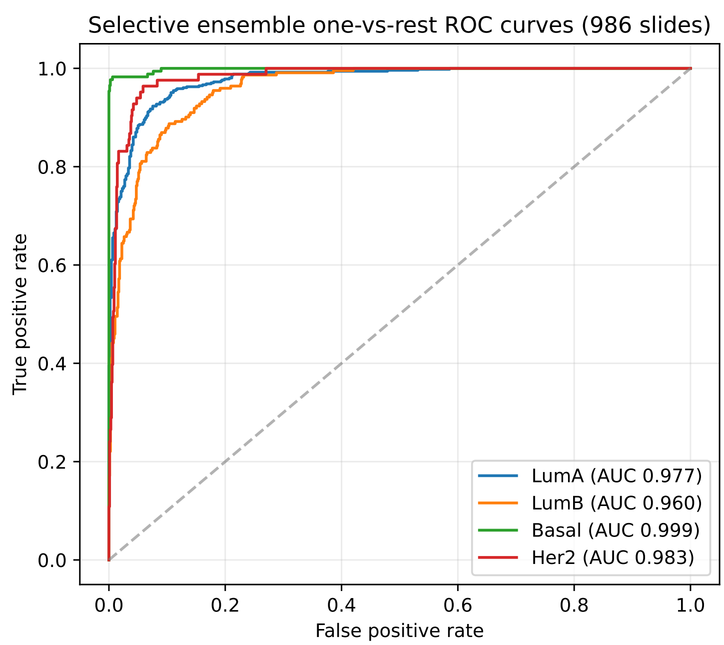

Figure 3: Selective ensemble one-vs-rest ROC curves on the 986-slide cohort.

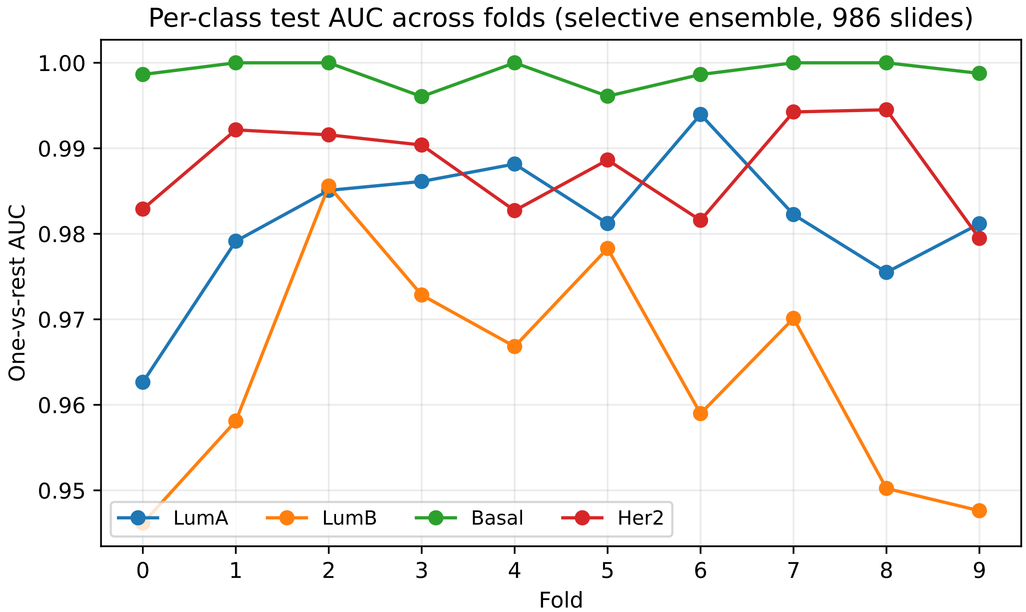

Figure 4: Selective ensemble per-class test AUC across folds. Basal-like sits at or near 1.0 in every fold; Luminal B is the most variable class.

4.4 Gated Fusion Behaviour

For the gated fusion model, the aggregate mean WSI gate is 0.4709 and the mean RNA gate is 0.5291. Mean WSI gates are similar across subtypes: 0.4661 for Luminal A, 0.4793 for Luminal B, 0.4739 for Basal-like, and 0.471 for HER2-enriched. Correct predictions have mean WSI gate 0.4703 and incorrect predictions 0.4757. The learned gate is a stable soft mixture that slightly favours RNA rather than a confidence switch driven by morphology.

5 Comparison to the WSI-only Branch

The WSI-only branch is the existing CLAM-MB + UNI2-h baseline. RNA-seq supplies a much stronger PAM50 signal than histology alone on this cohort. Figure 5 contrasts the WSI-only branch, the matched RNA branch, and the selective ensemble on the same 986-slide cohort.

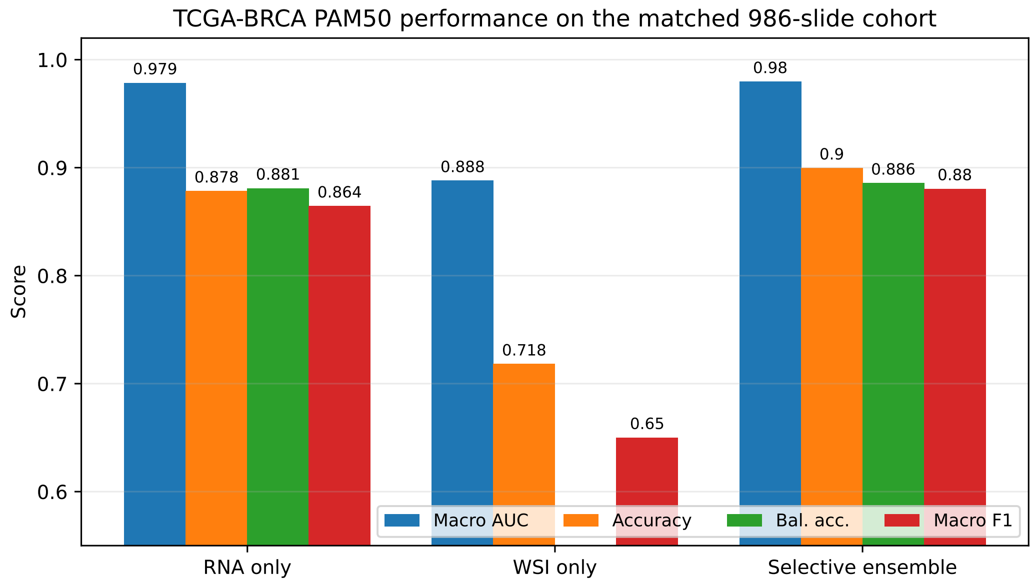

Figure 5: TCGA-BRCA PAM50 performance on the matched 986-slide cohort. The selective ensemble adds about 12 macro AUC points, 18 accuracy points, and 23 macro F1 points over WSI alone, and about 2 accuracy points and 1.6 macro F1 points over RNA alone.

The RNA branch is a strong molecular reference. WSI-only does not reach the RNA branch on any class-sensitive metric, but it does carry useful complementary signal: the selective ensemble gains over RNA-only come from the fusion components that include the WSI branch.

6 WSI–RNA Fusion Evaluation

The full set of fusion experiments is summarised in Table 4 and Table 5. Three observations:

- Concat and gated fusion add 1 accuracy point over RNA-only (0.8884 and 0.8874 vs. 0.8783) but trade 0.002–0.004 macro AUC. They reach RNA-only performance from a different decision surface.
- Late probability fusion matches the RNA macro AUC at 0.9789 with a small accuracy gain. Residual fusion is similar on AUC but loses 0.027 balanced accuracy because the WSI residual hurts minority-class recall.
- The selective ensemble is the only method that beats matched RNA on every metric in Table 4. The fold-weight summary shows that no single submodel dominates across folds: the validation-selected weight on residual is non-zero in every fold, while RNA-only weight is selected for several folds and concat is preferred when WSI–RNA alignment is strong.

7 Comparison to SOTA

Hezil et al. [1, 2] report a TCGA-BRCA PAM50 pipeline that combines CTransPath WSI features, RNA-seq features, multimodal fusion, and confidence-aware routing. Tables 7 and 8 compare their reported metrics to the 986-cohort results in this report. Cohort size, splits, feature extractors, and reporting protocol differ, so absolute numbers are not a direct ranking.

Table 7: Single-modality comparison on TCGA-BRCA PAM50 subtype classification. Best within each modality block is bolded. Literature numbers are taken from [2].

Method	Setting	Cohort	Acc. ↑	Macro F1 ↑	Weighted F1 ↑
Reference					
Hezil et al. [2]	RNA-seq only	1022 patients	**0.91**	**0.9**	**0.91**
Hezil et al. [2]	WSI only, CTransPath	1022 patients	0.62	0.57	0.64
This work, matched 986-slide cohort					
Matched RNA only	RNA MLP	986 slides	0.8783	0.8644	0.8799
WSI only	UNI2-h + CLAM-MB	986 slides	0.7181	0.65	0.7179

Table 8: Multimodal comparison on TCGA-BRCA PAM50 subtype classification. Best within each block is bolded.

Method	Setting	Cohort	Acc. ↑	Macro F1 ↑	Weighted F1 ↑
Reference					
Hezil et al. [2]	Multimodal concatenation	1022 patients	0.935	0.92	0.93
Hezil et al. [2]	Multimodal gated fusion	1022 patients	0.942	0.93	0.94
Hezil et al. [2]	Multimodal cross-attention	1022 patients	0.948	0.93	**0.95**
Hezil et al. [2]	Routing-based fusion	1022 patients	**0.9505**	**0.93**	**0.95**
Hezil et al. [1]	Routing-based fusion	924 patients	0.9493	–	–
This work, matched 986-slide cohort					
WSI + RNA concat	UNI2-h CLAM-MB + RNA MLP	986 slides	0.8884	0.8653	0.8896
WSI + RNA gated	UNI2-h CLAM-MB + RNA MLP	986 slides	0.8874	0.8667	0.8885
Late probability fusion	RNA MLP + UNI2-h CLAM-MB	986 slides	0.8803	0.8657	0.8812
WSI + RNA residual	Frozen RNA + residual WSI head	986 slides	0.8803	0.8575	0.8797
Selective ensemble	Calibrated RNA / WSI / concat / gated / residual	986 slides	**0.8996**	**0.8804**	**0.8999**

The Hezil et al. routing-based fusion remains roughly 5 accuracy points above our selective ensemble. Three structural reasons explain the gap. First, our cohort is restricted to 986 slides that have both UNI2-h features and matched RNA-seq, while the literature uses 1022 patients with a different inclusion rule. Second, our WSI branch uses CLAM-MB on UNI2-h slide features rather than CTransPath bag features as in the reference work. Third, our protocol concatenates ten patient-level held-out test folds and reports the resulting metrics on every slide; the reference protocol is not publicly identical.

The pattern across the two settings is consistent with prior multiomics work [4]: the RNA branch carries strong PAM50 signal and the routed multimodal fusion adds the largest incremental gain. Histology-first work [3] and recent multimodal histology–molecular fusion [5] support the same direction.

8 Discussion, Limitations, and Next Steps

8.1 Headline Result

On the matched 986-slide cohort, the strongest configuration is the calibrated selective ensemble: 0.8996 accuracy, 0.9796 macro AUC, 0.8856 balanced accuracy, 0.8804 macro F1, 0.8999 weighted F1, and 0.0256 expected calibration error. It is also the most stable across folds at 0.9829 ± 0.0053 macro AUC and 0.9001 ± 0.0248 accuracy. Aggregate gain over RNA-only is 2.1 accuracy points and 1.1 macro AUC points, and over WSI-only it is 18.2 accuracy points and 9.2 macro AUC points.

8.2 Where Errors Concentrate

The Luminal A vs. Luminal B border accounts for most of the remaining errors. 53 Luminal A slides are predicted as Luminal B and 32 Luminal B slides are predicted as Luminal A, which matches the biology: the two classes share hormone receptor status and differ mainly in proliferation. HER2-enriched recall is 0.83 with 83 supporting slides; the small support drives both the variance and the lower precision. Basal-like is effectively solved (0.998 AUC, 0.98 recall).

8.3 Limitations

- **Cohort size and source.** All experiments are on a single TCGA-BRCA cohort with 986 slides. Without an external validation set we cannot separate cohort-specific effects from model behaviour.
- **Class imbalance.** HER2-enriched has 83 slides, so any single misclassified slide shifts per-fold metrics noticeably. Reported standard deviations should be read with this in mind.
- **Slide-level matching.** Restricting to slides with matched RNA-seq removes RNA as a confounding source of leakage but reduces the WSI-only sample size from 1,040+ to 986. The WSI-only baseline reported here is therefore a lower bound on the achievable WSI-only number on the unconstrained cohort.
- **Fusion expressivity.** Concat and gated fusion never beat RNA on macro AUC, which suggests the joint head is not extracting genuinely new structure beyond what each modality already produces. The selective ensemble works because it routes between modalities per fold, not because it discovers a richer joint representation.
- **No external validator.** Macro AUC alone does not reflect calibration. Only the selective ensemble reports ECE because the other models are not temperature-calibrated. A routine calibration step on the other branches would make the comparison cleaner.

8.4 Next Steps

- **Stronger fusion head.** Cross-attention between RNA tokens and CLAM patch attention is the obvious next step. The Hezil et al. routing-based result is the target.
- **Confidence-aware routing.** The current selective ensemble uses validation-fixed weights. Replacing fold-fixed weights with a slide-level routing rule that uses RNA confidence to fall

back to WSI (or vice versa) is the most likely source of further gain on the Luminal A / Luminal B border.

- **Larger WSI cohort.** Refit WSI-only on the full unconstrained TCGA-BRCA WSI cohort to give a fair WSI baseline that does not pay the matched-cohort cost.
- **External validation.** Evaluate the trained selective ensemble on a held-out cohort with paired WSI and RNA-seq (e.g. CPTAC-BRCA) to test cohort transfer.
- **Calibration on every branch.** Apply the same temperature calibration to RNA, concat, gated, and residual outputs so reported ECE is comparable across rows.

References

[1] Hezil, N. et al. Selective multimodal deep learning for reliable breast cancer subtype classification from histopathology and genomic data. *Medical Engineering & Physics* (2025). PubMed record.

[2] Hezil, N. et al. Selective multimodal breast cancer subtype classification implementation and reported metrics. Code repository.

[3] Ektefaie, Y. et al. Integrative multiomics-histopathology analysis for breast cancer classification. *npj Breast Cancer* 7, 147 (2021). Article.

[4] Lin, Y. et al. Classifying Breast Cancer Subtypes Using Deep Neural Networks Based on Multi-Omics Data. *Genes* 11(8), 888 (2020). Article.

[5] Ben Rabah, C. et al. A Multimodal Deep Learning Model for the Classification of Breast Cancer Subtypes. *Diagnostics* 15(8), 995 (2025). Article.